Dengue Database — QA Report

File: **data/processed/dengue_cases_combined.xlsx** · Built by ml/clean_dengue.py · Report date: 18/10/2026

Verdict: is it ready for EARS?

Overall	**PASS** **Ready, with EARS settings.** Complete weekly calendar, nulls instead of gaps, a data_quality value on every row and a data_flag on suspect weeks. No data step is outstanding; what remains are settings in the EARS code (§5).
NATIONAL sheet	**PASS** **Ready.** 13 missing weeks and 14 year-end reporting artefacts are already flagged. EARS must skip any week whose data_quality is not “ok”.
LOCAL sheet	**WARN** **Ready, with flags.** 230 weeks of possible non-reporting (17 places) and 24 one-week spikes are marked in data_flag, values unchanged. The incomplete final week was removed. EARS must decide how to treat flagged weeks and use floors for small counts (§5).

1. Dataset overview

	NATIONAL	LOCAL
Rows	261	32,886
Series	1 (Philippines)	126 provinces / cities in 17 regions
Weeks	261 (10/01/2016 – 03/01/2021)	261 per place (same range)
Source	OpenDengue V1.3 (WHO WPRO / MOH)	DOH local-level dataset (HDX)
data_quality	ok 234, missing 13, incomplete 8, backlog 6	ok 31,910, suspect swap 665, missing 311
data_flag	none	possible non-reporting 230, one-week spike 24
doh_status	no baseline 167, normal 49, epidemic 40, unreliable week 4, alert 1	no baseline 19,537, normal 8,839, epidemic 2,995, alert 1,113, unreliable week 402

Columns: geographic_level, (region, geography_name — local only), period_start, value, data_quality, data_flag, epi_year, epi_week, alert_threshold, epidemic_threshold, doh_status, doh_note.

2. Checks passed

Check	Result	
No duplicate place-week rows	0 duplicates	**PASS**
Every date is a Sunday (week start)	all 33,147 rows	**PASS**
Complete weekly calendar, no skipped weeks	126 × 261 local, 261 national	**PASS**
Each place belongs to one region	0 conflicts	**PASS**
Values are non-negative whole numbers	0 negative, 0 non-integer	**PASS**
Null value ⇔ data_quality = missing	consistent on both sheets	**PASS**
Epi weeks numbered 1–52 (53 in 2020)	matches the MMWR calendar	**PASS**
DOH thresholds recomputed independently	11,436 rows, 0 mismatches	**PASS**
data_flag only on reported (“ok”) weeks	0 on non-ok rows	**PASS**
No blank doh_status cells	every row has a status + note where needed	**PASS**
National vs local totals agree	correlation 0.95; national ≈ 0.93× local sum	**PASS**

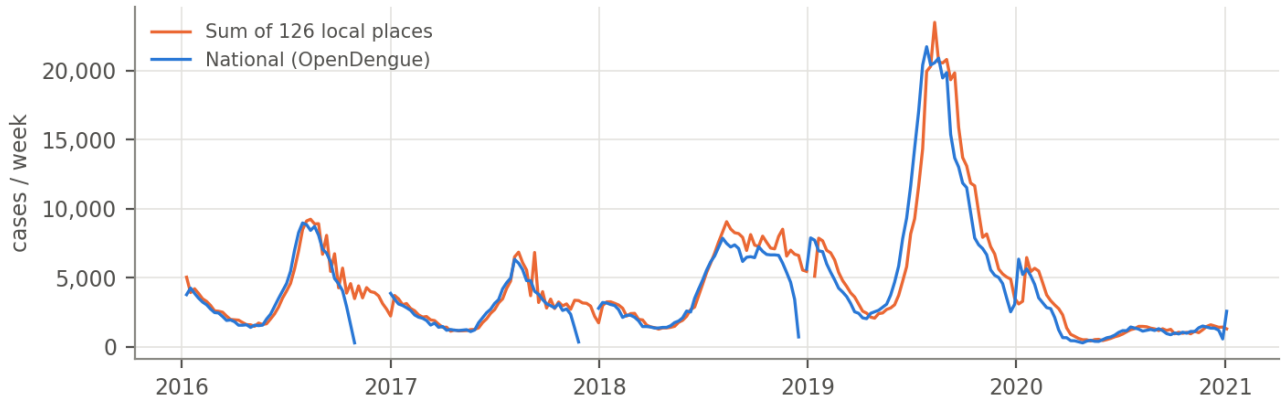


Figure 1. National series against the sum of all local places. They track closely except at year-end (national fades out, then jumps) and during the 2019 epidemic peak.

3. Findings

Ranked by how much they affect EARS. The status after the latest rebuild is shown on each finding.

HIGH F1. Local zeros that look like non-reporting — flagged

230 weeks in 17 places are runs of 4+ consecutive zeros in a place that normally reports 10+ cases a week. These are almost certainly “no report received”, recorded as 0 — the source does not distinguish the two. They cluster in recognisable events: **BARMM in 2017** (Lanao del Sur and Maguindanao during the Marawi siege), **Apr–Jun 2020** (COVID-19 lockdowns: NCR cities, Baguio, Angeles, Zambales), **Tacloban Jan–Sep 2016**, and **Eastern Visayas Nov 2020 – Jan 2021** (end of the series).

Can it be fixed? Not from this data: the true counts for these weeks were never recorded. Only DOH / the regional surveillance units hold corrected figures, so this is a reporting-system limitation rather than a cleaning error.

Why it matters: a zero inside the 7-week baseline drags the mean and SD down, so the first real week after the run looks like an outbreak. **Done:** these weeks have data_flag = “possible non-reporting” (rule: 4+ consecutive zeros where the place’s median non-zero week is 10+). Values are kept as published; doh_note repeats the flag.

Place	Region	Zero run	Weeks	Typical / wk
TACLOBAN	REGION VIII-EASTERN VISAYAS	10/01/2016 – 25/09/2016	38	10
MAGUINDANAO	BARMM	08/01/2017 – 14/05/2017	19	17
MAGUINDANAO	BARMM	27/08/2017 – 19/11/2017	13	17
MAGUINDANAO	BARMM	28/05/2017 – 13/08/2017	12	17
NORTHERN SAMAR	REGION VIII-EASTERN VISAYAS	01/11/2020 – 03/01/2021	10	10
LANAO DEL SUR	BARMM	05/11/2017 – 31/12/2017	9	16
TACLOBAN	REGION VIII-EASTERN VISAYAS	09/10/2016 – 27/11/2016	8	10
LANAO DEL SUR	BARMM	08/01/2017 – 26/02/2017	8	16
ZAMBALES	REGION III-CENTRAL LUZON	05/04/2020 – 17/05/2020	7	13
TAGUIG CITY	NATIONAL CAPITAL REGION	12/04/2020 – 24/05/2020	7	15
LANAO DEL SUR	BARMM	09/07/2017 – 20/08/2017	7	16
BAGUIO CITY	CAR	12/04/2020 – 24/05/2020	7	10

Top 12 of 30 runs shown.

HIGH F2. Local one-week spikes — flagged

24 weeks are more than 8× the place’s 9-week rolling median and at least 4× both of their neighbours: the count jumps for a single week and falls straight back (e.g. Cebu: 0 → 2,123 → 1 in Sep 2017). Real outbreaks build and decline over several weeks, so these are most likely backlogs dumped into one week or entry errors. Even the spikes inside the 2019 epidemic (Zamboanga City 298 → 3,382 → 377, Antique 209 → 2,036 → 196) have this shape. A short manual review against DOH bulletins is advised.

Can it be fixed? Not reliably: there is no record of which earlier weeks the cases belong to, so spreading them back would be a guess. **Done:** these weeks have data_flag = “one-week spike”, values kept. If a review against DOH bulletins confirms one as real, its flag can be removed.

Place	Week	Week before	Cases	Week after	9-wk median	× larger neighbour
ZAMBOANGA CITY	28/07/2019	298	3,382	377	280	9.0×
ANTIQUE	11/08/2019	209	2,036	196	196	9.7×
CAGAYAN DE ORO CITY	23/09/2018	104	998	131	109	7.6×
BOHOL	11/09/2016	125	934	83	86	7.5×
CAGAYAN DE ORO CITY	18/11/2018	94	818	94	94	8.7×
LANAO DEL NORTE	11/11/2018	73	686	54	66	9.4×
ANTIQUE	13/08/2017	36	678	72	30	9.4×
CEBU CITY	06/08/2017	60	673	59	57	11.2×
ILOCOS NORTE	10/11/2019	78	652	44	78	8.4×
LA UNION	02/12/2018	24	551	33	30	16.7×
NEGROS ORIENTAL	06/08/2017	58	546	62	60	8.8×
BUKIDNON	06/08/2017	46	542	60	57	9.0×
LA UNION	18/11/2018	30	335	24	35	11.2×
OCCIDENTAL MINDORO	08/07/2018	13	333	39	35	8.5×
PUERTO PRINCESA CITY	30/06/2019	19	332	42	19	7.9×
MANDAUE CITY	17/02/2019	42	322	28	28	7.7×
TANDAG CITY	19/03/2017	5	229	3	3	45.8×
ZAMBALES	10/11/2019	15	221	17	20	13.0×
ISABELA	09/02/2020	29	199	11	14	6.9×
ROMBLON	17/11/2019	4	121	0	4	30.2×
PATEROS	19/03/2017	1	94	0	0	94.0×
PATEROS	04/06/2017	0	86	0	0	86.0×
DAVAO DEL NORTE	30/07/2017	7	67	1	4	9.6×
LAPU-LAPU CITY	15/05/2016	2	36	1	2	18.0×

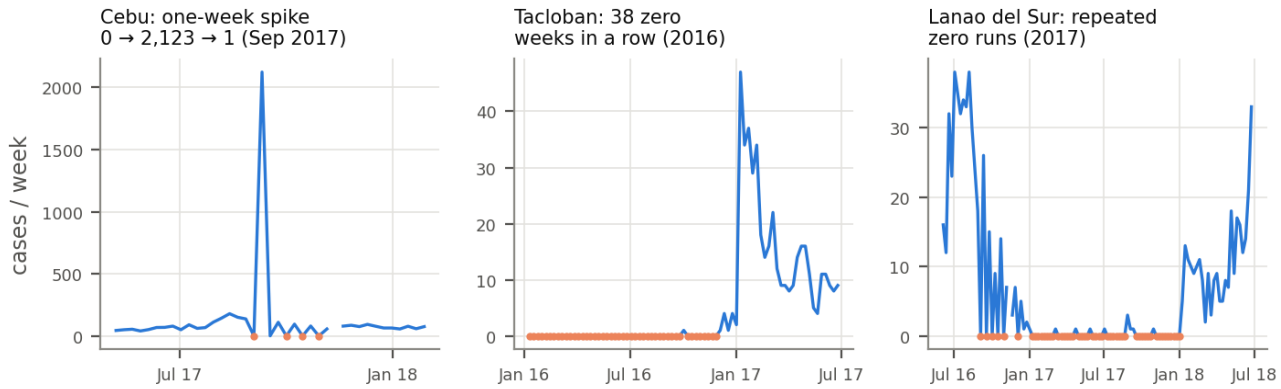


Figure 2. Examples of F1 and F2. Orange dots are zero weeks.

MEDIUM F3. Local final week was incomplete — removed

The local data’s last week (10/01/2021) totalled 784 cases against about 1,384 in the weeks before: it was extracted before most of its reports had arrived. **Done:** both sheets now end on 03/01/2021, so national and local still cover the same 261 weeks.

MEDIUM F4. Small counts make thresholds hair-trigger

Places with very few cases produce baselines with little or no spread: 121 rows have a baseline SD of 0 (alert = epidemic threshold, so any case above the mean is an “epidemic”), and 157 local weeks are marked

“epidemic” with 5 cases or fewer — mostly Davao Occidental, Tawi-Tawi, Sulu, Catanduanes and Dinagat Islands. 9 places are zero in over a quarter of weeks (Batanes 69%).

Not a data error — this is how the data is. It is handled in the EARS settings (§5): a minimum SD and a minimum case count before an alarm is raised.

LOW F5. Davao Occidental is patchy

54 missing weeks scattered through 2016–2018 and an average of 1 case a week. EARS will rarely have 5 usable baseline weeks there before 2019; expect mostly blank results.

INFO F6. Cities and provinces listed side by side

Highly urbanised cities appear as their own rows (e.g. CEBU and CEBU CITY). The local sum is close to the national total, which suggests provinces exclude their cities, but this is not documented. Do not add a city to its province without confirming.

INFO F7. National year-end artefacts — already handled

Each year national fades out before a December gap (“incomplete”, 8 weeks) and jumps in early January (“backlog”, 6 weeks). These are flagged and excluded from DOH thresholds.

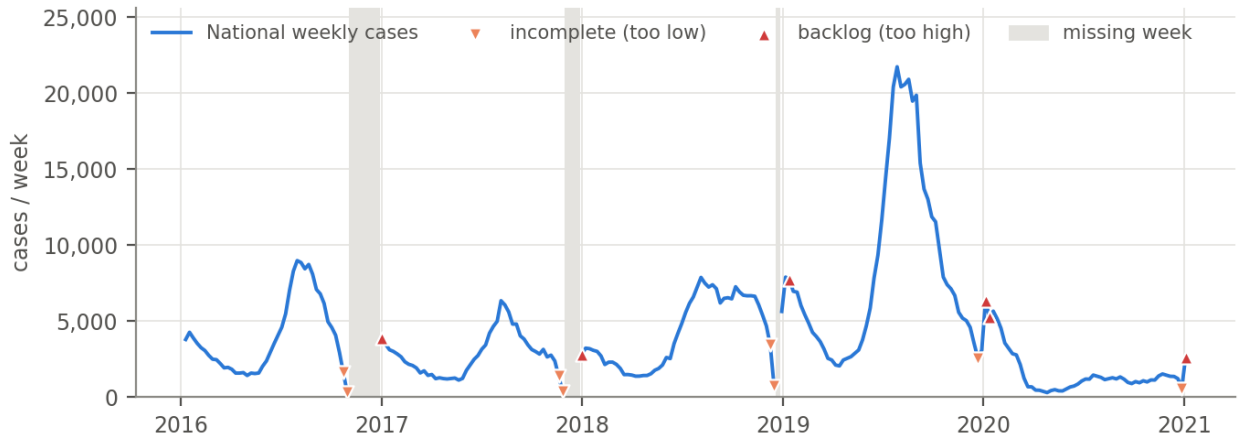


Figure 3. National series with its data_quality flags. Grey bands are missing weeks.

4. Cleaning already applied

Step	Detail
Trimmed to a common range	10/01/2016 – 03/01/2021: the local data’s span minus its incomplete final week; national 2012–2015 and 2021–2023 dropped
Complete weekly calendar	every series has all 261 weeks; missing weeks kept as null rows, never deleted, zero-filled or interpolated
Dropped columns	national: everything except geographic_level, period_start, value; local: deaths
Removed non-data rows	national title/description rows; local HXL tag row (#adm2+name …)
Name fixes	BASILIG CITY → BISLIG CITY, NUEVA VISCAYA → NUEVA VIZCAYA, SISQUIJOR → SIQUIJOR; leading spaces trimmed
Region fixes	REGION VII-EASTERN VISAYAS → REGION VIII-EASTERN VISAYAS; REGION IVB → IV-B; case unified
data_quality flag	ok / missing on both sheets; incomplete / backlog on national year-end weeks
data_flag	local: “possible non-reporting” (F1) and “one-week spike” (F2); values unchanged
DOH thresholds	Per DOH PIDSR Manual of Procedures (2014) §5.3.5, over the same epi week in 3 prior years: alert = mean + 1·SD, epidemic = mean + 2·SD (sample SD); 2-of-3 fallback; week 53 uses week 52; every deviation explained in doh_note

5. EARS readiness checklist

Requirement	National	Local
One row per week, no skipped weeks	**PASS**	**PASS**
Missing weeks are null, not deleted or zero	**PASS**	**PASS**
Unreliable weeks flagged so baselines can skip them	**PASS**	**PASS** — F1, F2 in data_flag; F3 removed
Enough history for a 7-week baseline (C1–C3)	**PASS**	**PASS** (except Davao Occidental)
Counts large enough for a stable SD	**PASS**	**WARN** — F4, set floors

Before running EARS

Data: nothing outstanding. Optional: review the 26 F2 spikes against DOH bulletins.

EARS settings (in the EARS code):

1. Use only weeks with data_quality = “ok”; any other week is treated as null.
2. Weeks with a data_flag still get their own EARS result, with the flag repeated in the note. Leave them out of other weeks’ baselines (recommended), or keep them in and note every result whose baseline contains one.
3. Compute a baseline only when at least 5 of its 7 weeks are usable (the agreed middle-ground rule); otherwise no result.
4. No result when the current week is not “ok”; C3 needs C2 at t, t−1 and t−2.
5. Set a minimum SD (e.g. 1 case) so a flat baseline cannot divide by zero, and require a minimum count (e.g. 5 cases) before a week can alarm, to stop single-case alarms in small places (F4).

Limitation for the paper: F1 and F2 reflect late or missing reporting at source. EARS can only be as timely and complete as the reports it receives.